Ahmed Islam Farouk Abbas Badr

Undergraduate Artificial Intelligence Science Student
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Summary

Highly motivated Artificial Intelligence student with hands-on research internship experience at J**ames Madison University, Virginia Tech, and the University of Louisville.** Proven expertise in developing and deploying complex systems in Computer Vision (YOLO, ROS2), NLP (LLMs, RAG), and medical AI, earning a 3rd place research award at R!L. Recipient of the Presidential Medal of Excellence for athletic achievements.

Education

Computer Science and Engineering | Alamein International University | Major in Artificial Intelligence **Sep. 2022–Jun. 2026**

- **CGPA: 3.73 / 4.0**

High School Graduate | El Zahraa Language School **Graduated - 2022**

Skills

- **Programming Languages:** Python, Java, JavaScript, C++, R, TypeScript, SQL, NOSQL
- **AI Domains & Core ML:**Machine Learning, Neural Networks, Deep Learning, Computer Vision, NLP, Generative AI, Data Preprocessing & Augmentation, Feature Extraction, Transfer Learning, Optimization Algorithms
- **Computer Vision:** OpenCV, ROS2, ZED SDK,Facial Expression Recognition, Object Detection (YOLO, Faster R-CNN), Image Processing
- **AI Frameworks & Libraries:** PyTorch, TensorFlow/Keras, Scikit-learn, Pandas, NumPy
- **Development Tools:** Docker, Streamlit, Flask/FastAPI, Git, GitHub, Linux, Bash, Jupyter, VSCode
- **Soft Skills**: Leadership, Teamwork, Time Management, Problem-Solving, Adaptability and Communication
- **Languages:** English (Fluent) | Arabic (Native)

Work Experience

Research Intern | James Madison University | Harrisonburg, VA, USA | On-site **Jun. 2025 – Jul. 2025**

- Collaborated with JMU students under **Dr. Samy El-Tawab** and **Dr. Ahmad Salman** on computer vision and AI
- Developed autonomous golf cart navigation system using object detection with YOLO algorithms and ZED 2i stereo camera, implementing ROS2 framework on Linux with Docker containerization
- Built sobriety detection system using Siamese networks for user profiling and eye-tracking algorithms to detect intoxication levels through facial recognition and behavioral pattern analysis

Software Engineering Intern | Virginia Tech | Blacksburg, VA, USA | Remote **Jan. 2025 – Jun. 2025**

- Engaging in an advanced training program with Virginia Tech students under **Dr. Mohamed Seyam** to develop LLM-based AI search platform using BERT, LLMs and vector databases
- Building full-stack application with React.js, Node.js, PostgreSQL and RESTful APIs with microservices architecture
- Creating AI-powered search functionality with NLP, vector embeddings, computer vision OCR for diagram-based search and conversational AI chatbots

Research Assistant | Alamein International University **Jan. 2025 – Present**

- **Conducting Research** with **Dr. Mohamed Abdelrahman** on developing **Speech Emotion Recognition (SER) and Facial Emotion Recognition (FER)** systems, leveraging deep learning architectures and multimodal fusion for robust classification.
- Collaborating with **Dr. Islam Elgedawy** on research about **Retrieval-Augmented Generation (RAG) pipelines**, integrating vector databases and embedding-based search to enhance large-language-model reasoning and knowledge retrieval.

Research Intern | University of Louisville | Louisville, KY, USA | On-site **Jul. 2024 – Sep. 2024**

- Collaborated with **research team** under **Dr. Ayman El-Baz** to develop AI diagnostic models using Python, TensorFlow, scikit-learn for non-invasive renal rejection detection in kidney transplant patients
- Performed data preprocessing, feature engineering, statistical analysis on medical imaging, genomic, and clinical datasets using Pandas, NumPy, OpenCV

- Implemented machine learning algorithms (Random Forest, SVM, Neural Networks, XGBoost, Cat Boost) with cross-validation and hyperparameter tuning
- In recognition of my work I achieved **Third Place** in the Prestigious (R!L) competition for research excellence Presenting a Poster for this project and now a Remote volunteer in Bioengineering Labs

English Assistant Teacher | Egyptian American Cultural Centre-Alexandria **Jul. 2021 - Sep. 2021**

- Assisting the head English teacher in class activities, lessons, and language

Taekwondo Coach | Smouha Sporting Club Alexandria **Nov. 2021 - Present**

- Working closely with the head coach and others to develop and implement training session plans

Most Notable Projects

AI Based approach to classify renal rejection grades | UOFL Research Project (3rd place)

- An AI Based non-invasive approach to classify renal rejection grades in kidney transplant patients in the bioengineering labs at the University of Louisville where I spent summer at the university working on the project integrating imaging, genomic and clinical data to classify renal rejection into grades comparing each combination of classifications and presenting a poster at R!L **winning third** in the process!

Autonomous Golf Cart Navigation | James Madison University

- Built ROS 2–based perception and control system for autonomous navigation using ZED 2i stereo vision, YOLOv8/SSD detection, and depth estimation. Designed modular publisher–subscriber nodes for planning and collision avoidance with sub-100ms latency. Integrated actuator interfaces for sensor calibration, motor control, and safety protocols.

Sobriety Detection System | James Madison University

- Designed Developed a real-time sobriety testing platform using Siamese networks, gaze-tracking, and facial-behavior analysis. Built a modular pipeline with OpenCV, OpenFace (288+ features), and CSV-based data export for analysis.

Intelligent Projects Assets Repository | Virginia Tech Capstone Project

- Designed scalable microservices backend for LLM-powered semantic search with FastAPI, and vector databases (FAISS). Implemented Sentence-BERT embeddings, multimodal retrieval (text, diagrams, voice), and RAG pipelines.

Interactive Egyptian Museum Application | 2024 Hackathon (3rd place)

- Created a museum application with an interactive chatbot and a chatroom for the 2024 hackathon **winning third** in the process

Snake Game with AI and Multiplayer Mode

- Developed an advanced multiplayer Snake game featuring AI-driven opponents and real-time gameplay. Implemented reinforcement learning algorithms to optimize AI behavior and improve decision-making. Utilized threading and concurrency to ensure smooth real-time interactions. Integrated networking components to enable competitive multiplayer experience.

SkyVision | Graduation Project (Ongoing) | Collaboration With James Madison University

- Developing an AI-powered aerial monitoring and vision system for real-time environmental and infrastructure analysis using drone swarms under joint mentorship from faculty members at AIU and James Madison University, and in collaboration with JMU research students. Implementing ROS2-based multi-drone control, computer vision pipelines, and collision avoidance with modular perception and planning nodes.Integrating LLM-assisted situational analysis for automated decision support and anomaly detection.

Publications & Research

SkyVision: A Real-Time Sensor Fusion Pipeline for Autonomous Drone Swarm Navigation **in Disaster** Response.

- ***Status: In Early Preparation (Graduation Project)***

A RAG-Based Multimodal Search Engine for Intelligent Project Asset Retrieval.

- ***Status: In Preparation***

Multimodal Emotion Recognition: Integrating Speech and Facial Expression Analysis for Enhanced Computing

- ***Status: In Preparation***

Courses

Successfully completed over **25+ professional courses** in AI, software engineering, and related fields
details available on my LinkedIn profile or upon request.

Extracurricular Activities

Startup Initiatives | Nomeda | Founder **Oct. 2024 - Present**

- As the founder of Nomeda I am leading multiple innovative ventures that integrate AI and software engineering.

IEEE Club Member - AI Committee **Dec. 2024 - Present**

- Actively engaged in IEEE events, workshops, and research communities, contributing to discussions on AI, machine learning, and software engineering

ICPC Club Member **Nov. 2023 - Present**

- Solving algorithmic and data structure challenges in a competitive programming environment, demonstrating strong problem-solving and coding skills under pressure.

Professional Taekwondo Athlete & Coach

- Third-Dan black belt and Egyptian National Team member with 43 medals (17 Gold, 10 Silver, 16 Bronze)
- **First Place** at African Games (Mar. 2024) | Medal of Excellence from the **Egyptian President** (Jul. 2024)

References Available on request